

Name; inshal nasir

Section : A

Roll no : 38

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam create-group --group-name SoftwareEngineering
{
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPAZCN5ZS5DAB4XH7OPZ",
    "Arn": "arn:aws:iam::623705167686:group/SoftwareEngineering",
  }
}
```

Q1.create group

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam get-group --group-name SoftwareEngineering
{
  "Users": [],
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPAZCN5ZS5DAB4XH7OPZ",
    "Arn": "arn:aws:iam::623705167686:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:41:35+00:00"
  }
}
```

q1_group_details

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam create-user --user-name Inshal
{
  "User": {
    "Path": "/",
    "UserName": "Inshal",
    "UserId": "AIDAZCN5ZS5DGDMQX5D6O",
    "Arn": "arn:aws:iam::623705167686:user/Inshal",
    "CreateDate": "2026-01-19T07:45:21+00:00"
  }
}
```

q1-create-user

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam get-user
--user-name Inshal
{
  "User": {
    "Path": "/",
    "UserName": "Inshal",
    "UserId": "AIDAZCN5ZS5DGDMQX5D60",
    "Arn": "arn:aws:iam::623705167686:user/Inshal",
    "CreateDate": "2026-01-19T07:45:21+00:00"
  }
}
```

q1_user_details

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam add-user
-to-group --user-name Inshal --group-name SoftwareEngineering
```

Q1_add_user_to_group

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam get-group
--group-name SoftwareEngineering
{
  "Users": [
    {
      "Path": "/",
      "UserName": "Inshal",
      "UserId": "AIDAZCN5ZS5DGDMQX5D60",
      "Arn": "arn:aws:iam::623705167686:user/Inshal",
      "CreateDate": "2026-01-19T07:45:21+00:00"
    }
  ],
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPAZCN5ZS5DAB4XH70PZ",
    "Arn": "arn:aws:iam::623705167686:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:41:35+00:00"
  }
}
```

q1_group_membership.png

```
• @23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam list-pol  
  icies --scope AWS --query "Policies[?PolicyName=='AdministratorAccess']"  
  [  
    {  
      "PolicyName": "AdministratorAccess",  
      "PolicyId": "ANPAIWMBCKSKIEE64ZLYK",  
      "Arn": "arn:aws:iam::aws:policy/AdministratorAccess",  
      "Path": "/",  
      "DefaultVersionId": "v1",  
      "AttachmentCount": 1,  
      "PermissionsBoundaryUsageCount": 0,  
      "IsAttachable": true,  
      "CreateDate": "2015-02-06T18:39:46+00:00",  
      "UpdateDate": "2015-02-06T18:39:46+00:00"  
    }  
  ]
```

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam attach-g  
  roup-policy --group-name SoftwareEngineering --policy-arn arn:aws:iam::aws:  
  policy/AdministratorAccess
```

```
• @23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam list-att  
  ached-group-policies --group-name SoftwareEngineering  
  {  
    "AttachedPolicies": [  
      {  
        "PolicyName": "AdministratorAccess",  
        "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"  
      }  
    ]  
  }  
• @23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $
```

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/inshal/section=permissions

Identity and Access Management (IAM)

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Dashboard

Access Management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analyzer
- Unusual access

Inshal

Summary

ARN: [arn:aws:iam::623705167686:user/inshal](#)

Console access: Disabled

Access key 1: [Create access key](#)

Created: January 18, 2026, 25:45 (UTC-08:00)

Last console sign-in: -

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name	Type	Attached via
AdministratorAccess	AWS managed - job function	Group SoftwareEngineering

Permissions boundary (not set)

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us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/policies/details/arn%3Aaws%3Aiam%3A%3Aaws%3Apolicy%2FAdministratorAccess...

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AdministratorAccess

Provides full access to AWS services and resources.

Policy details

Type: AWS managed - job function

Creation time: February 06, 2015, 10:59 (UTC-08:00)

Edited time: February 06, 2015, 10:59 (UTC-08:00)

ARN: [arn:aws:iam::aws:policy/AdministratorAccess](#)

Permissions Entities attached Policy versions (1) Last Accessed

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Initial | IAM | Global x Initial | IAM | Global x SoftwareEngineering | IAM | Global x Create access key | IAM | Global x

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups/details/SoftwareEngineering?section=users

Identity and Access Management (IAM)

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SoftwareEngineering

Summary

User group name: SoftwareEngineering

Creation time: January 18, 2025, 23:41 (UTC-08:00)

ARN: arn:aws:iam::625705167686:group/SoftwareEngineering

Users (1)

Permissions Access Advisor

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☐	User name	Groups	Last activity	Creation time
☐	Initial	1	None	23 minutes ago

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Question 2

```
provider "aws" {
  region = "me-central-1"
  profile = "default"
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet" {
  vpc_id          = aws_vpc.myapp_vpc.id
  cidr_block      = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}
```

```
resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.myapp_vpc.id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_route_table" "rt" {
  vpc_id = aws_vpc.myapp_vpc.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }
  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

resource "aws_route_table_association" "rta" {
  subnet_id      = aws_subnet.myapp_subnet.id
  route_table_id = aws_route_table.rt.id
}
```

```
resource "aws_route_table_association" "rta" {
  subnet_id      = aws_subnet.myapp_subnet.id
  route_table_id = aws_route_table.rt.id
}

data "http" "my_ip" {
  url = "https://icanhazip.com"
}

locals {
  my_ip = "${chomp(data.http.my_ip.body)}/32"
}
```

```
resource "aws_security_group" "default_sg" {
  vpc_id = aws_vpc.myapp_vpc.id
  name   = "${var.env_prefix}-default-sg"

  ingress {
    description = "SSH from my IP"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [local.my_ip]
  }

  ingress {
    description = "HTTP from anywhere"
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTPS from anywhere"
```

⏏ Help ⏏ Write Out ⏏ Where To ⏏ Out ⏏ Execute

```
resource "aws_key_pair" "serverkey" {
  key_name   = "serverkey"
  public_key = file("/home/codespace/.ssh/id_ed25519.pub")
}
```



```
GNU nano 7.2          terrafrom.tfvars *
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix           = "dev"
instance_type        = "t3.micro"
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38-/terraform_lab (main) $ terraform apply
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami                  = "ami-0f2b4fc905b0bd1f1"
  + arn                  = (known after apply)
  + associate_public_ip_address = true
```

Question 3

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38-/terraform_lab (main) $ mkdir -p Lab_exam/ansible
cd Lab_exam/ansible
@23-22412-038-sys → .../-inshal-nasir-38-/terraform_lab/Lab_exam/ansible (main)
```

[frontend]

<frontend-public-ip>

[backends]

<backend1-public-ip>

<backend2-public-ip>

<backend3-public-ip>

[all:vars]

ansible_user=ec2-user

ansible_ssh_private_key_file=~/.ssh/id_ed25519

GNU nano 7.2

playbook.yml *

name: Configure EC2 Web Server

hosts: ec2

become: true

tasks:

- name: Update all packages

yum:

name: "*"

state: latest

- name: Stop Nginx if running

service:

name: nginx

state: stopped

enabled: no

ignore errors: yes

